

TABLE I
LITERATURE-GROUNDED STREAMING AND RESIDENT-WRITE CAPABILITIES OF REPRESENTATIVE CIM IMPLEMENTATIONS

Technology / implementation	Mode; logical $n_{\text{out}} \times n_{\text{in}}$; input/weight	ρ [GB/s]	τ	RI*	Evidence / key condition
SRAM / Jia et al. [1]	ACIM; 64×1152 ; 4b/4b	2.9	—	—	M/C; 4 input phases; 20 MS/s ADCs.
SRAM / D6CIM [2]	DCIM; 16×128 ; 8b/8b	0.72	—	—	M/C; 64 clocks/MVM at 360 MHz.
DRAM / Ambit [3]	In-array Boolean row logic	—	—	—	S/D; 49 ns AAP; no full MVM schedule.
Gain-cell eDRAM / Xie [4]	ACIM; 1×27 ; 8b/8b	1.4	—	—	M/C; one MAV group, 4 phases / 20 ns.
3D NAND / Shim–Yu [5]	ACIM model; 16×4608 ; 8b/8b	—	—	—	S/D; 303 ns WL setup only; verify untimed.
2D NOR Flash / Guo [6], [7]	Analog cascade; $784 \rightarrow 64 \rightarrow 10$	—	—	—	M/D; $< 1 \mu\text{s}$ classification; preload excluded.
RRAM / NeuRRAM [8]	ACIM; 256×128 ; 4b/analog	0.016	—	—	M/C; $3.9 \mu\text{s}$; $56 \mu\text{s}$ /cell write is projected.
MRAM / Jung [9]	ACIM; 64×64 ; bipolar 1b/1b	0.089	0.044	2	M/C; row write: 64 weights in 2 clocks.
PCM / 64-core chip [10]	ACIM; 256×256 ; 8b/analog	0.49	—	—	S/C; measured-chip RTL, 4 phases / 520 ns.
FeRAM / C2FeRAM [11]	Proposed ACIM; discrete 2T2C	—	—	—	D/D; device tests; no complete timed macro.
3D FeNOR / Zhou 2025 [12]	Device multiply; binary state	—	—	—	D/D; $+4.25/-3.25 \text{ V}$, 50 ns; RAW delay $< 100 \text{ ns}$.
Vertical FeFET / Zhou 2026 [13]	Memory array; binary state	—	—	—	D/D; $\pm 2 \text{ V}$, 20 ns; 1 Kib patterns verified.

Scope: Native local service boundaries; one core/group unless specified. Packed logical input bytes are counted once per complete stated-precision evaluation; physical bit slices and complementary cells are not extra payload. Analog outputs retain the reported quantization/error contract. $\text{RI}^* = \rho/\tau$ only for a compatible pair; GB/s is decimal. These configurations are not normalized for process, area, capacity or power.

Evidence: First code: M, measured chip; D, measured device; S, simulation. Second code: D, directly reported parameter; C, converted capability. PCM timing is RTL-derived on a measured-chip architecture. — means insufficient evidence for this service boundary, not impossibility. AAP is a DRAM row-operation primitive; RAW means read-after-write. FeNOR switching pulses and RAW delays do not specify CIM throughput. Full conditions, write/verify evidence and unavailable timing are retained in the supporting records.

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